

**Inter Pipeline Propylene Ltd.
Propane Dehydrogenation Project**



**E&I Building Cable Tray Supports
PDH-00-ST-CAL-0017**



This document has been revised as indicated below. Please destroy all previous revisions.

Rev	Date	Revision Description	PREP	CHKD	APP'D	MANAGEMENT	
						Fluor	IPPL
0	30-JUL-2018	Issued for Information – Project 1202	CW	JR	-	-	-

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PROJECT: Propane Dehydrogenation Project
Location: Strathcona County, Alberta
Title: E&I Building Cable Tray Supports
Calculation No.: PDH-00-ST-CAL-0017
By: Coltin Walsh

Contact No.: A6KV
Date: 30-Jul-2018

Check Level: 2B
Chk'd: JR

Strees Log:

Stress Load Review Status:

			Date
a)	Stress loads have been assumed in line with FD guideline	☐ Check	
b)	Stress loads has been assumed in line with FD guideline for design and compared with Stress ISOs at a later date (Stress walk-through report is attached in attachment #)	☐ Check	
c)	Stress ISOs have been used for producing the calculation (Stress loads are attached in Appendix B)	☐ Check	
d)	No pipes supported on this structure	☒ Check	13-Jul-18

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1.0 DESIGN SUMMARY

1.1 Purpose of Calculation

The purpose of this calculation is to design the E&I building cable tray supports as per KIEWIT-SITE-RFI-000183.

1.2 Scope of Calculation

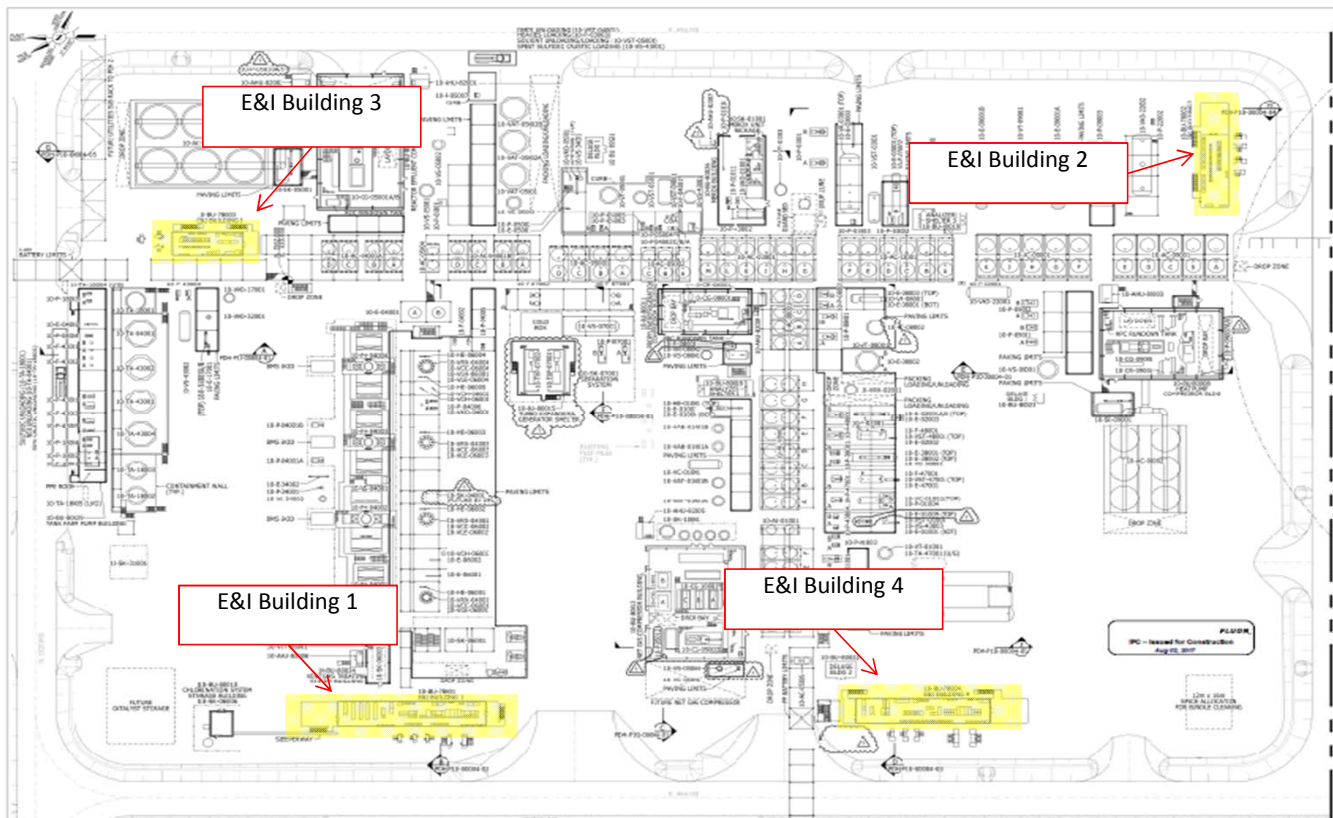
This calculation report includes the following items:

1. Standard load development
2. Structural steel member design

1.3 Structure Layout and Location

The cable tray support beams are located beneath the E&I Buildings and connected to the piles. Connection of beam to piles are simple shear connections, with one end slotted to facilitate construction and prevent tying the piles together.

There are 4 E&I buildings located in areas 01CC, 02PF, 02SP, and 01RE/03PU:

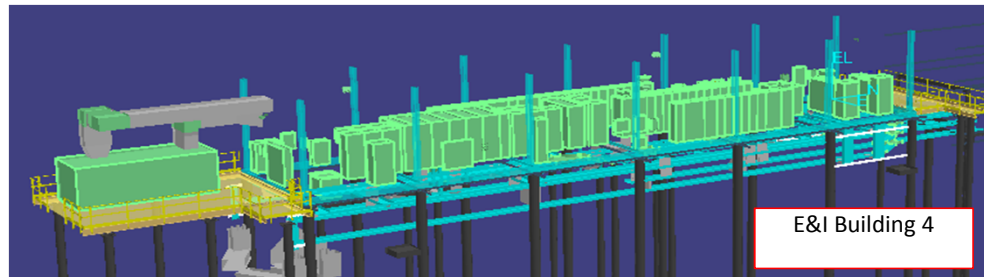
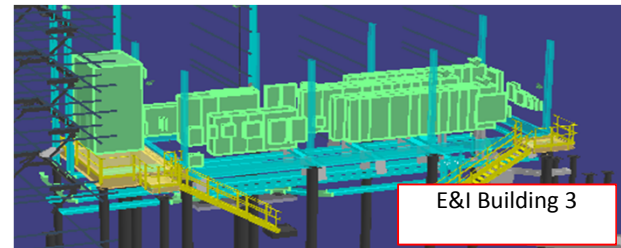
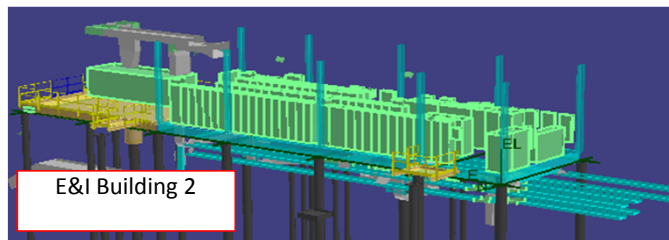
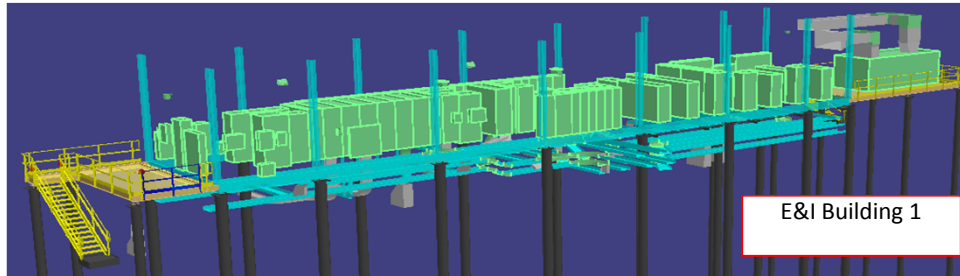




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S3D Screenshots:



1.4 Method of Analysis

Structural steel calculations and analysis were performed manually in Excel and using RISA 3D.

1.5 Design Criteria

Design criteria is based on specification PDH-00-ST-SPC-0001 - Structural Engineering Design Criteria.

1.5.1 General

A. Climatic Data

Site meteorological and geological data for the facility is as follows:

Ground Snow Load, S_s	=	1.6 kPa
Ground Rain Load, S_r	=	0.1 kPa

B. Importance Factors

As per Jdb. 2.20, the site will have importance factor: **HIGH**.

Load	ULS	SLS
Snow, I_s	1.15	0.90



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1.5.2 Governing Codes

The following codes were used for analysis and design:

NBC 2010	National Building Code of Canada 2010
ABC 2014	Alberta Building Code 2014
CSA S16-09	Design of Steel Structures

1.5.3 Specifications and Guidelines

The following specifications and guidelines were used for analysis and design:

Specifications:

Jdb. 2.20 - Structural Engineering Design Criteria
 Propane Dehydrogenation Project specifications and standards

1.5.4 Strength and Serviceability Requirements

The structure was designed based on industry limits and the requirements specified in the above specifications.

1.5.5 Materials

See standard drawing PDH-SD20-00001 for steel material.

1.6 Software

Microsoft Excel - for calculations and organization of calculation report
 RISA 3D - for structural analysis and steel design

1.6.3 Electronic Files

File Name	Description	Directory
PDH-01-ST-CAL-0017.pdf	Calculation report	215_STRUCTURAL\215E_ENGINEERING_DESI GN_DATA\215E_014_R_Proj_Calculations_Sketc hes\PDH-00-ST-CAL-0017_0000-SS-E&I Building Cable Tray Supports\Excel Working
PDH-01-ST-CAL-0017.pdf	Calculation report	215_STRUCTURAL\215E_ENGINEERING_DESI GN_DATA\215E_014_R_Proj_Calculations_Sketc hes\PDH-00-ST-CAL-0017_0000-SS-E&I Building Cable Tray Supports\Compiled Calculation
E&I Building Tray Supports.r3d	RISA Model	215_STRUCTURAL\215E_ENGINEERING_DESI GN_DATA\215E_014_R_Proj_Calculations_Sketc hes\PDH-00-ST-CAL-0017_0000-SS-E&I Building Cable Tray Supports\RISA Model
E&I Building Tray Supports.rcn	RISACconnection	215_STRUCTURAL\215E_ENGINEERING_DESI GN_DATA\215E_014_R_Proj_Calculations_Sketc hes\PDH-00-ST-CAL-0017_0000-SS-E&I Building Cable Tray Supports\RISACconnection

2.0 UNCERTAINTIES AND RISKS

Uncertainty	Mitigation	Remaining Risk	Comment	Resolution
N/A				

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3.0 STRUCTURAL DESCRIPTION AND LAOD DEVELOPMENT

3.1 KIEWIT-SITE-RFI-000183 - RFI Description

Subject*: ELECTRICAL - CABLE TRAY - E&I Building Tray Support Proposal

Message*:

Per Fluor design cable tray is to be hung off of the E&I building's steel, which will not be arriving on site until the following year. Kiewit would like to proceed with cable tray installation prior to their arrival.

Discipline*: EL - Electrical

Area*: 0000 - General Document

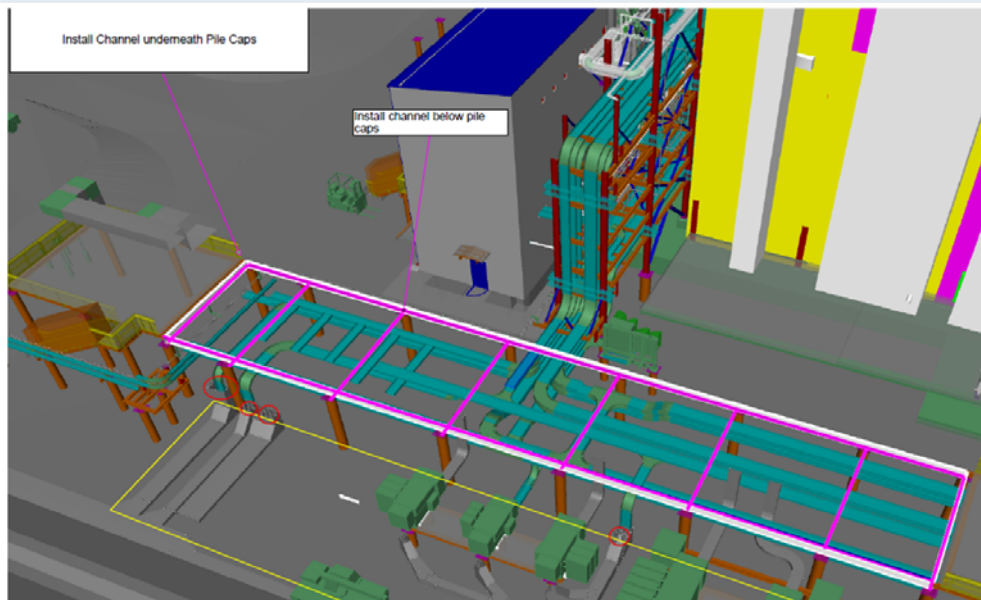
Location*: 00 - Common / Onplot

RFI Type*: SITE - Site RFI

Proposed Solution*:

Kiewit proposes to install C100x8, as per attached drawing (E&I Building 4 Markup), between pilings, to facilitate cable tray installation prior to arrival of E&I buildings. Channel is proposed to be welded below to the pile caps. Tray supports to remain the same.

Fluor to confirm if the use of C100x8 channel and standard electrical weld detail is acceptable. Design will be typical for all 4 of the E&I buildings.





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As per KIEWIT-SITE-RFI-000183, Kiewit proposes to support the cable trays below the E&I buildings with beams spanning between the piles to expedite the construction activities. Originally, the cable trays were to be suspended from the underside of the building skid after building installation. However, due to the building arriving 1 - 2 years after start of construction, Kiewit proposes to install primary cable tray runs prior to building arrival and installation. The section size proposed by Kiewit (C100) is not adequate for the 7m span by inspection. After discussions with IPPL and Kiewit, Fluor CSA is to proceed with the design of these members and connections.

FLUOR 200 - Path forward for electrical trays on piles to E-houses
Bin Li to: Thomas.charlton, Coltin Walsh, Rana Pankajakshan, ben.santos... 07/06/2018 12:02 AM
Cc: bill.king, Elzbieta Szaro, Richard Foulkes, Shamina Hakda, CA A6KV Mail-in... [Hide Details](#)

From: Bin Li/CA/FL/FluorCorp [Sort List...](#)
To: Thomas.charlton@kiewit.com, Coltin Walsh/CA/FL/FluorCorp@FluorCorp, Rana Pankajakshan/CA/FL/FluorCorp@FluorCorp, ben.santos@interpipeline.com, chad.bemis@kiewit.com
Cc: bill.king@interpipeline.com, Elzbieta Szaro/CA/FL/FluorCorp@FluorCorp, Richard Foulkes/CA/FL/FluorCorp@FluorCorp, Shamina Hakda/CA/FL/FluorCorp@FluorCorp, CA A6KV Mail-in/FluorCorp@FluorCorp

Yesterday we had a meeting and discussed about issues and decisions for electrical trays on piles to all E-houses.

1. The original construction plan for PDH project was to install U/G cable after E-houses are delivered and installed, the design was completed based on this plan. The Kiewit construction plan is to install the U/G power cable prior to E-house installation. This will avoid idling of U/G cable installation until E-house arrives and open up construction work front now and increase work efficiency later. Per new plan and given that the piles around E-houses are installed, Kiewit suggest to place electrical trays on new steels required to be placed between piles and below the pile caps so that the U/G cables can be brought up to above ground and laid on the trays in front of E-houses, eventually link to the waterfall trays from the buildings.

2. Fluor expressed concern of not meeting electrical code requirement for the clearance left between the top of the tray and bottom of the pile cap per the steel previously selected. Through the meeting discussion, a new size of the steel has been selected and confirmed that it will meet the clearance requirement per code and the stiffness requirement for the load estimated from both cable, tray and snow.

3. Per IPPL's request, Fluor will provide the design of the steel and connection details to the piles. Kiewit will purchase the steel and install them at site. Fluor will produce official structural drawings instead of sketches for the design. Once the steel is sized and dimension is confirmed, but prior to issuing the drawings, Fluor will inform Kiewit for the purchase of the steel to avoid the delay of the supply. Fluor electrical engineering group confirmed that this approach will not change the original MTO of the tray outside of the E-houses and will instruct procurement to purchase the tray immediately to avoid the delay of the supply. It is also confirmed that this approach will not change current electrical design. Fluor electrical will provide support to structural steel design by CSA group.

4. In addition to the discussion of this solution, Kiewit requested clarification to the 'missing' steel at E-house for the cable tray support. Fluor clarified the details to be followed and explained why the steels are not shown in the model and should not be considered 'missing' because E-house supplier will provide the drawings to show these steels. Based on the current situation of the PO with E-house vendor, it is anticipated that the drawings may be available in couple of weeks. Fluor is to expedite the vendor drawings.

Bin

Bin Li, P.Eng | **FLUOR CANADA LTD.** | Engineering Management | bin.li@fluor.com | IODC 40.1068 | O +1.403.538.1068

3.2 Model Construction

Due to multiple different cable tray loads and winding tray layout, it is difficult to determine critical members by inspection. Therefore, the beam grid was modelled and analyzed using RISA 3D.

Geometry:

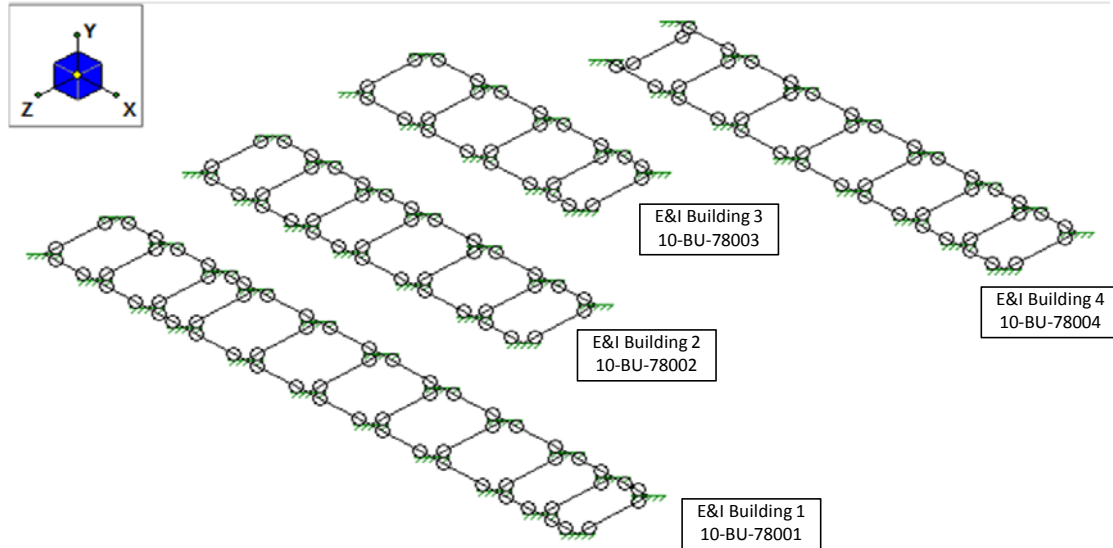
$L_{Trib} = 6 \text{ m}$: Tributary length between piles
 $L_{Beam} = 6.985 \text{ m}$: Center-to-center spacing of pile

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The RISAs model was developed as a grid. The Pile support points were taken as Fixed supports, and the longitudinal and transverse beams were pin-connected to the supports.



3.3 Load Development

3.3.1 - BLC 1: Self Weight (SW):

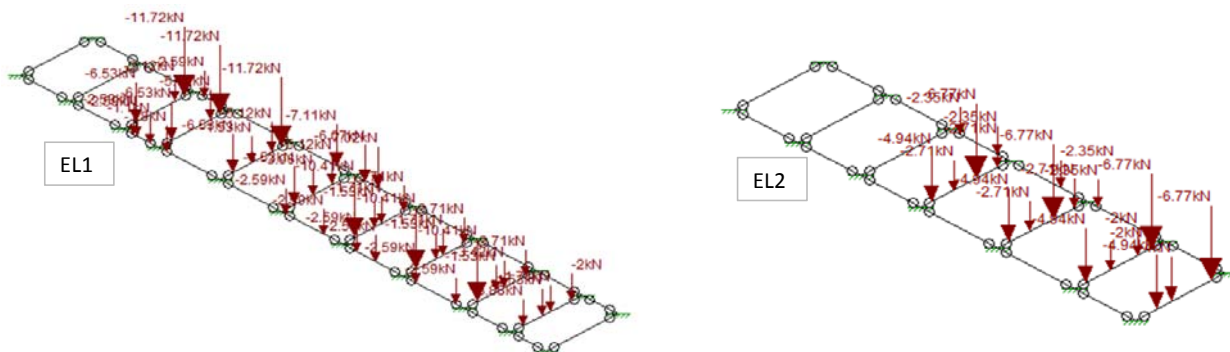
Dead load is applied automatically in RISA using the gravity loading feature. A factor of 1.15 is used for connection allowance.

3.3.2 - BLC 2 - 5: Electrical Cable Tray (EL):

Exact cable tray loading was provided by Electrical. Cable layout was taken from S3D model: Longitudinal tray has tributary length of 6m. Transverse tray has an assumed tributary length of 4m. See Appendix A for detailed cable tray loading calculations.

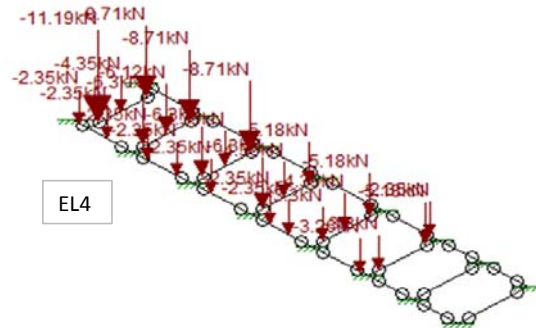
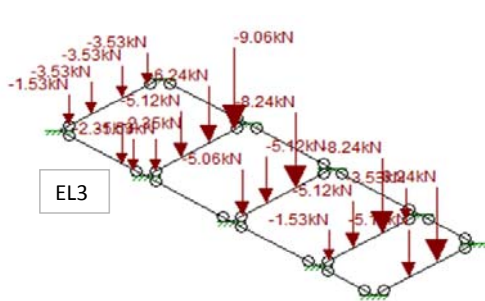
Tray loads are applied as point loads in the RISA model at the **center of tray location**.

Trays that are stacked have loads added and applied at the same RISA node.



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3.3.3 - BLC 6 - 9: Snow Load (SL):

Snow load is calculated using the above climatic data and the ABC 2014:

$$S = I_s [S_s (C_b C_w C_s C_a) + S_r] \quad : \text{ABC 2014}$$

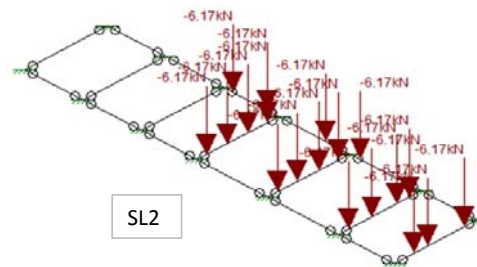
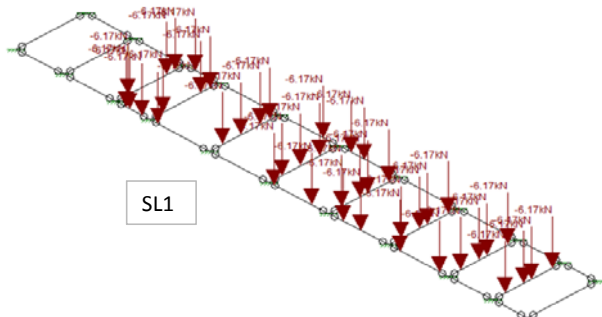
I_s	=	1.15	: Importance factor for ULS, High
I_s	=	0.90	: Importance factor for SLS, High
S_s	=	1.60 kPa	
S_r	=	0.10 kPa	< $S_s (C_b C_w C_s C_a) = 1.28$
C_b	=	0.80	: Basic roof snow load; 0.8 for small roofs
C_w	=	1.00	: Wind exposure factor, 1.0 for exposed
C_s	=	1.00	: Slope factor, 1.0 for flat surfaces
C_a	=	1.00	: Shape factor, assumed 1.0

Blanket Snow Load:	SL_{SLS}	=	1.24 kPa	>	1.0 kPa
	SL_{ULS}	=	1.59 kPa		

Snow load is applied at each cable tray location, using the width of tray = 0.650m:

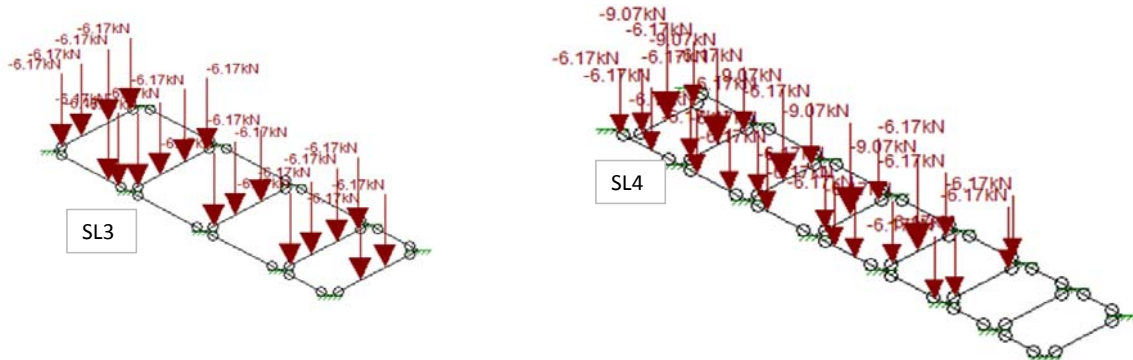
SL	=	$SL_{ULS} * 0.648m * 6m$	: 650 mm tray
	=	6.17 kN	
SL	=	$SL_{ULS} * 0.9525m * 6m$	: 950 mm tray
	=	9.07 kN	

For stacked trays, SL is applied for the top tray only, assuming the other trays are shielded.



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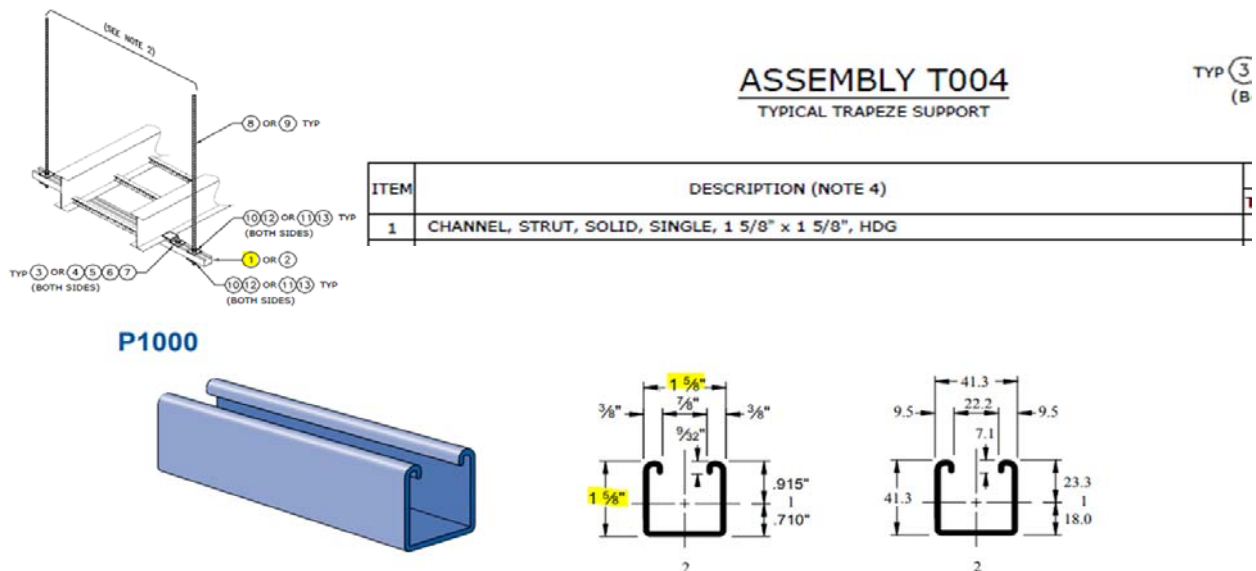
4.0 LOAD COMBINATIONS

The Cable Tray Load case was increased by 10% to account for uncertainties and weight of unistruts.

3D Load Combinations												
Combinations Design												
	Description	So...	PDelta	SRSS	BLC	Factor	BLC	Factor	BLC	Factor	BLC	Factor
2	EL	☒	Y		2	1.1	3	1.1	4	1.1	5	1.1

Typically used P1000 unistrut only has dead load of 283 kg / 100m or approximately $2.83 \text{ kg/m} \times 9.81 / 1000 \times 1.2\text{m} = 0.033\text{kN}$ per tray.

Minimum tray load = 1.18 kN: $0.033 / 1.18 = 3\%$; Therefore, 10% increase is conservative.



Pierced channels are found on pages 60 and 61.

Weight: 190 Lbs/C Ft (283 kg/100 m)

A total of 2 governing ULS and SLS load combinations were used. First combination assumes no snow load, second combination assumes full snow load.



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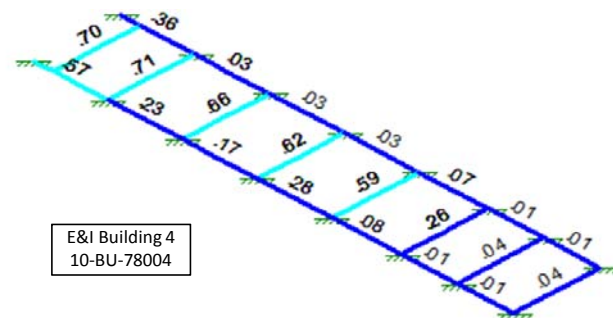
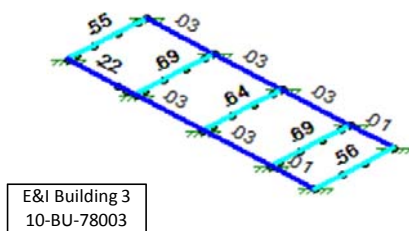
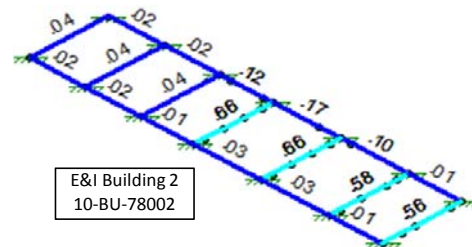
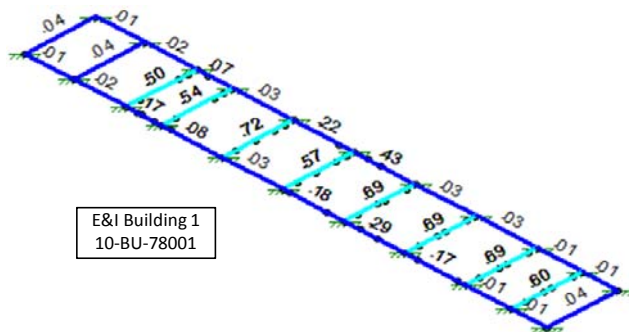
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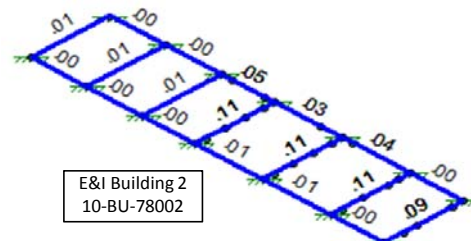
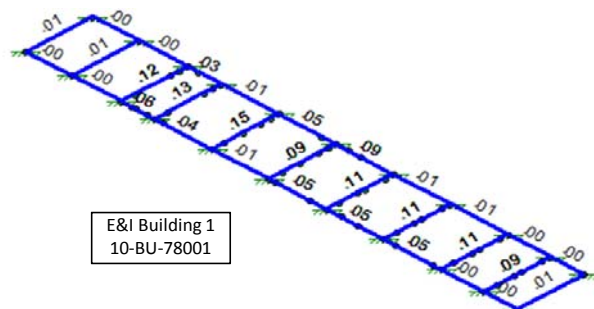
Load Combinations												
Combinations		Design										
	Description	So...	PDelta	S	BLC	Factor	BLC	Factor	BLC	Factor	BLC	Factor
1	SW	☐	Y		1	1						
2	EL	☐	Y		2	1.1	3	1.1	4	1.1	5	1.1
3	SL	☐	Y		6	1	7	1	8	1	9	1
4	ULS (1.4SW + 1.4EL)	☒	Y		L1	1.4	L2	1.4				
5	ULS (1.25SW + 1.25EL + 1.5SL)	☒	Y		L1	1.25	L2	1.25	L3	1.5		
6	SLS (SW + EL)	☒	Y		L1	1	L2	1				
7	SLS (SW + EL + 0.78SL)	☒	Y		L1	1	L2	1	L3	.78		

5.0 STRUCTURAL DESIGN AND RESULTS:

Bending Unity Check:

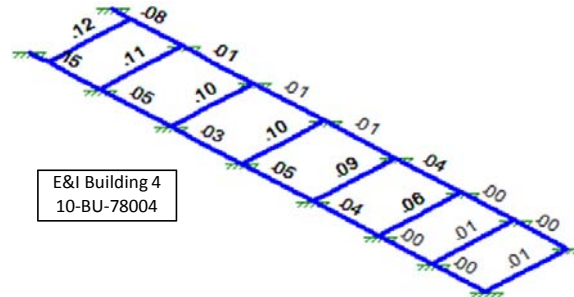
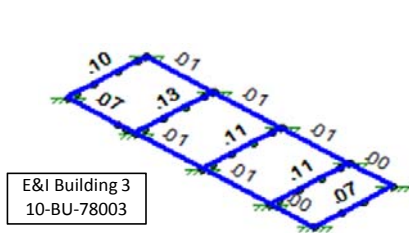


Shear Unity Check:



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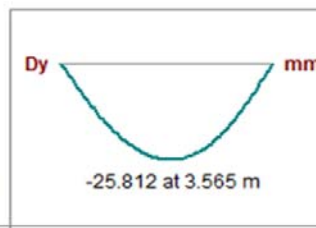


Maximum Bending Unity = 0.72
 Maximum Shear Unity = 0.15.
 Therefore, member unity check is OK!

Deflection Check:

30 Member Section Deflections									
	LC	Member Label	S...	x [mm]	y [mm]	z [mm]	x Rotat...	(n) L/y Ratio	(n) L/z Ratio
1	7	M5	1	0	0	0	0	NC	NC
2			2	0	-18.102	0	0	385.817	NC
3			3	0	-25.794	0	0	270.765	NC
4			4	0	-19.301	0	0	361.84	NC
5			5	0	0	0	0	NC	NC

Beam: **M5**
 Shape: **W200x42**
 Material: **G40.21 350W**
 Length: **6.984 m**
 I Joint: **N10**
 J Joint: **N9**
 LC 7: **SLS (SW + EL + 0.78SL)**
 Code Check: **0.465 (bending)**
 Report Based On 97 Sections



Maximum beam deflection = 26mm; L / 270.
 Deflection is OK.

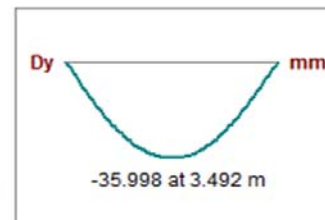
Member M6 in E&I Building 1 had to be reduced from W200x42 to W150x37 to maintain electrical code clearances. Under service loads, deflection of this member increases to 36mm (L / 190) while bending utilization increases to 0.73. As per discussion with A. Sharif (P.Eng), deflection of this member is acceptable as the system is flexible and snow load will not be present once the building is in place. Therefore, W150x37 is OK.

Bending

Beam: **M6**
 Shape: **W150x37**
 Material: **G40.21 350W**
 Length: **6.984 m**
 I Joint: **N12**
 J Joint: **N11**
 LC 5: **ULS (1.25SW + 1.25EL + 1.5SL)**
 Code Check: **0.729 (bending)**
 Report Based On 97 Sections

Deflection

Beam: **M6**
 Shape: **W150x37**
 Material: **G40.21 350W**
 Length: **6.984 m**
 I Joint: **N12**
 J Joint: **N11**
 LC 7: **SLS (SW + EL + 0.78SL)**
 Code Check: **0.460 (bending)**
 Report Based On 97 Sections



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Connection Design:

Waiward will be responsible for connection design. Therefore, a force set is provided:

3D Member Section Forces						
	LC	Member Label	S...	Axial[kN]	y Shea...	z Shea...
1	5	M5	1	0	30.643	0
2			2	0	-44.726	0
3	5	M80	1	0	31.762	0
4			2	0	-39.551	0
5	5	M4	1	0	24.489	0
6			2	0	-39.522	0

Callout	Fx	Fy	Fz	Mx	My	Mz	Notes
V1	20	50	5	-	-	-	
V2	-	50	5	-	-	-	Axial Slot = 21mm x 70mm

A long slot is provided on one end of the beam, allowing for $70/2 - 19.05/2 = \pm 25$ mm construction tolerance.

Axial Force from Wind = $1.15 * 0.43 * 0.9 * 2.0 * 2.0 * 0.160\text{m} * 6\text{m} * 1.4 * 8$ cable trays = ~20 kN. (Enveloped load)

Based on Waiwards currently provided connection details, similar detail to BS-51 is checked assuming 1 - row of bolts only:

20 kN Axial Tension

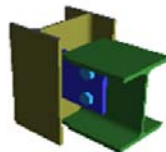


Beam to Pile

CSA

Column/Beam Shear Tab Shear Connection

Connection design



Material Properties:

Column	W200x36	G40.2-350W	$F_y = 350.00$ MPa	$F_u = 450.00$ MPa
Beam	W200x42	G40.2-350W	$F_y = 350.00$ MPa	$F_u = 450.00$ MPa
Plate	P12.70x140.00x145.00	G40.2-350W	$F_y = 350.00$ MPa	$F_u = 450.00$ MPa

Input Data:

Shear Load	50.00 kN	User Input Shear Load
Axial Load	-20.00 kN	User Input Axial Force (tension)

Note: Unless specified, all code references are from CSA S16-09

 Collapse All

 Expand All

Limit State	Required	Available	Unity Check	Result
Geometry Restrictions at Beam				FAIL
Column Weld Limitations				PASS
Rotational Ductility				FAIL
Beam Shear Yield	53.85 kN	306.86 kN	0.18	PASS
Plate Shear Yield	50.00 kN	382.85 kN	0.13	PASS
Beam Shear Rupture	53.85 kN	265.68 kN	0.20	PASS
Plate Shear Rupture at Beam	50.00 kN	331.47 kN	0.15	PASS
Plate Axial Yield	20.00 kN	580.07 kN	0.03	PASS
Plate Tension Rupture at Beam	20.00 kN	415.77 kN	0.05	PASS
Beam Block Shear	53.85 kN	232.15 kN	0.23	PASS
Plate Block Shear	53.85 kN	336.33 kN	0.16	PASS
Plate Tearout on Plate at Beam	20.00 kN	424.34 kN	0.05	PASS

PROJECT: Propane Dehydrogenation Project
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Date: 30-Jul-2018

Check Level: 2B
Chk'd: JR

Lateral Stability / Stabilizer Plates	53.85 kN	962.35 kN	0.06	PASS
Plate Flexural Yield			0.08	PASS
Plate Flexural Rupture			0.15	PASS
Plate Flexural Buckling	50.00 kN	107.02 kN	0.47	PASS
Bolt Bearing on Beam	53.85 kN	158.47 kN	0.34	PASS
Bolt Bearing on Plate at Beam	53.85 kN	158.47 kN	0.34	PASS
Weld at Column	1.51 kN/mm	1.87 kN/mm	0.81	PASS

20 kN Axial Compression

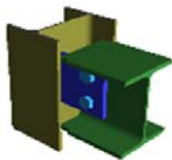


Beam to Pile

CSA

Column/Beam Shear Tab Shear Connection

Connection design



Material Properties:

	Column	Beam	Plate
W200x36	G40.2-350W	G40.2-350W	G40.2-350W
F _y = 350.00 MPa	F _y = 350.00 MPa	F _y = 350.00 MPa	F _y = 350.00 MPa
F _u = 450.00 MPa	F _u = 450.00 MPa	F _u = 450.00 MPa	F _u = 450.00 MPa

Input Data:

Shear Load	50.00 kN	User Input Shear Load
Axial Load	20.00 kN	User Input Axial Force (compression)

Note: Unless specified, all code references are from CSA S16-09

Collapse All
 Expand All

Limit State	Required	Available	Unity Check	Result
Geometry Restrictions at Beam				FAIL
Column Weld Limitations				PASS
Rotational Ductility				FAIL
Beam Shear Yield	53.85 kN	306.86 kN	0.18	PASS
Plate Shear Yield	50.00 kN	382.85 kN	0.13	PASS
Beam Shear Rupture	53.85 kN	265.68 kN	0.20	PASS
Plate Shear Rupture at Beam	50.00 kN	331.47 kN	0.15	PASS
Plate Axial Yield	20.00 kN	580.07 kN	0.03	PASS
Beam Block Shear	50.00 kN	232.15 kN	0.22	PASS
Plate Block Shear	50.00 kN	336.33 kN	0.15	PASS
Compression Buckling of the Plate	20.00 kN	556.34 kN	0.04	PASS
Lateral Stability / Stabilizer Plates	53.85 kN	962.35 kN	0.06	PASS
Plate Flexural Yield			0.09	PASS
Plate Flexural Rupture			0.13	PASS
Plate Flexural Buckling			0.50	PASS
Bolt Bearing on Beam	53.85 kN	158.47 kN	0.34	PASS
Bolt Bearing on Plate at Beam	53.85 kN	158.47 kN	0.34	PASS
Weld at Column	1.29 kN/mm	1.87 kN/mm	0.69	PASS

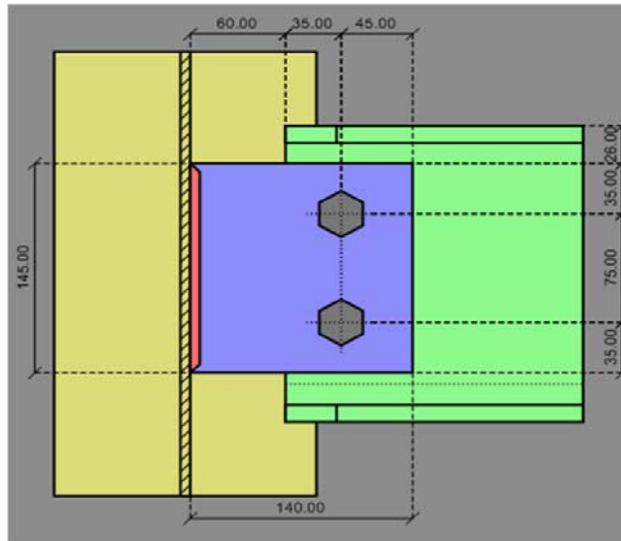
Connection Unity = 0.81



PROJECT: Propane Dehydrogenation Project
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Date: 30-Jul-2018
Check Level: 2B
Chk'd: JR

The only failures noted in the connection design include maximum plate thickness and rotational ductility; which can be waived with engineering judgement. Therefore, connection design is feasible. Actual connection detail will be completed by Waiward.



Beam location = 35mm gap + 25mm slot tolerance = 60 mm from face of column
 Assumed 35 mm edge distance.



PROJECT: Propane Dehydrogenation Project
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Date: 30-Jul-2018
Check Level: 2B
Chk'd: JR

Appendix A - Cable Tray Loading -



PROJECT: Propane Dehydrogenation Project
Location: Strathcona County, Alberta
Title: E&I Building Cable Tray Supports
Calculation No.: PDH-00-ST-CAL-0017
By: Coltin Walsh

Contact No.: A6KV
Date: 30-Jul-2018
Check Level: 2B
Chk'd: JR

BLDG NO	TRAY TAG	TRAY LOAD - kg/m (EXCL. SNOW AND WIND LOAD)	Tray Load [kN / m]	EL for Trib L 6m [kN]	EL for Trib L 4m [kN]
10-BU-78001	10-T2-CC002-60	26	0.26	1.53	1.02
	10-T2-RS001-60	29	0.28	1.71	1.14
	10-T3-RS001-60	29	0.28	1.71	1.14
	10-T3-RS002-60	34	0.33	2.00	1.33
	10-T3-RS003-60	66	0.65	3.88	2.59
	10-T3-CC003-60	111	1.09	6.53	4.36
	10-T5-CC011-60	44	0.43	2.59	1.73
	10-T5-CC012-60	65	0.64	3.83	2.55
	10-T5-CC013-60	90	0.88	5.30	3.53
	10-T5-RS001-60	53	0.52	3.12	2.08
	10-T5-RS002-60	34	0.33	2.00	1.33
	Other	60	0.59	3.53	2.35
	Total	641			

Member	Node	Tray Tag	EL [kN]
M3	N23	10-T3-CC003-60	-6.53
M3	N44	10-T5-RS001-60	-3.12
M3	N33	10-T5-CC011-60	-2.59
M3	N33	10-T5-CC012-60	-3.83
M3	N33	10-T5-CC013-60	-5.30
M4	N24	10-T3-CC003-60	-6.53
M4	N45	10-T5-RS001-60	-3.12
M4	N45	10-T5-RS002-60	-2.00
M4	N34	10-T5-CC011-60	-2.59
M4	N34	10-T5-CC012-60	-3.83
M4	N34	10-T5-CC013-60	-5.30
M5	N25	10-T3-CC003-60	-6.53
M5	N57	10-T2-CC002-60	-1.53
M5	N46	10-T5-RS001-60	-3.12
M5	N46	10-T5-RS002-60	-2.00
M5	N35	10-T5-CC011-60	-2.59
M5	N35	10-T5-CC012-60	-3.83
M5	N35	10-T5-CC013-60	-5.30
M6	N26	10-T3-CC003-60	-6.53
M6	N58	10-T2-CC002-60	-1.53
M6	N47	10-T5-RS001-60	-3.12
M6	N47	10-T5-RS002-60	-2.00
M7	N27	10-T3-RS003-60	-3.88
M7	N27	10-T3-CC003-60	-6.53
M7	N59	10-T2-CC002-60	-1.53
M7	N64	10-T2-RS001-60	-1.71
M7	N37	10-T3-RS002-60	-2.00
M7	N37	10-T3-RS001-60	-1.71
M8	N28	10-T3-RS003-60	-3.88
M8	N28	10-T3-CC003-60	-6.53
M8	N60	10-T2-CC002-60	-1.53
M8	N65	10-T2-RS001-60	-1.71
M8	N38	10-T3-RS002-60	-2.00



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Date: 30-Jul-2018
Check Level: 2B
Chk'd: JR

M8	N38	10-T3-RS001-60	-1.71
M9	N29	10-T3-RS003-60	-3.88
M9	N29	10-T3-CC003-60	-6.53
M9	N61	10-T2-CC002-60	-1.53
M9	N66	10-T2-RS001-60	-1.71
M9	N39	10-T3-RS002-60	-2.00
M9	N39	10-T3-RS001-60	-1.71
M10	N30	10-T3-RS003-60	-3.88
M10	N62	10-T2-CC002-60	-1.53
M10	N67	10-T2-RS001-60	-1.71
M10	N40	10-T3-RS002-60	-2.00

Longitudinal Struts

M14	N64A	Other	-2.35
M14	N65A	Other	-2.35
M14	N66A	10-T5-RS002-60	-1.33
M15	N67A	10-T2-CC002-60	-1.02
M17	N76	Other	-2.35
M17	N75	Other	-2.35
M18	N77	Other	-2.35
M18	N78	Other	-2.35
M19	N79	Other	-2.35
M19	N80	Other	-2.35
M24	N67B	Other	-2.35
M26	N72	10-T5-CC011-60	-1.73
M26	N72	10-T5-CC012-60	-2.55
M26	N72	10-T5-CC013-60	-3.53
M27	N73	10-T2-CC002-60	-1.02
M27	N73	10-T5-RS002-60	-1.33
M27	N73	10-T5-RS001-60	-2.08
M27	N73	10-T3-RS003-60	-2.59
M27	N74	10-T2-RS001-60	-1.14
M27	N74	10-T3-RS002-60	-1.33
M27	N74	10-T3-RS001-60	-1.14
M27	N74	10-T3-CC003-60	-4.36



PROJECT: Propane Dehydrogenation Project
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By: Coltin Walsh

Contact No.: A6KV
Date: 30-Jul-2018
Check Level: 2B
Chk'd: JR

BLDG NO	TRAY TAG	TRAY LOAD - kg/m (EXCL. SNOW AND WIND LOAD)	Tray Load [kN / m]	EL for Trib L 6m [kN]	EL for Trib L 4m [kN]
10-BU-78002	10-T3-NS001-60	60	0.59	3.53	2.35
	10-T3-NS002-60	24	0.24	1.41	0.94
	10-T3-NS003-60	70	0.69	4.12	2.75
	10-T3-NS004-60	45	0.44	2.65	1.77
	10-T3-SP001-60	46	0.45	2.71	1.81
	10-T3-SP002-60	46	0.45	2.71	1.81
	10-T5-NS001-60	34	0.33	2.00	1.33
	10-T5-NS002-60	34	0.33	2.00	1.33
	Other	60	0.59	3.53	2.35
	Total	419			

Member	Node	Tray Tag	EL [kN]
M63	N145	10-T3-NS002-60	-1.41
M63	N145	10-T3-NS001-60	-3.53
M63	N158	10-T3-SP001-60	-2.71
M63	N162A	10-T3-NS004-60	-2.65
M63	N162A	10-T3-NS003-60	-4.12
M63	N166A	10-T3-SP002-60	-2.71
M64	N146	10-T3-NS002-60	-1.41
M64	N146	10-T3-NS001-60	-3.53
M64	N159	10-T3-SP001-60	-2.71
M64	N163A	10-T3-NS004-60	-2.65
M64	N163A	10-T3-NS003-60	-4.12
M64	N167A	10-T3-SP002-60	-2.71
M65	N147	10-T3-NS002-60	-1.41
M65	N147	10-T3-NS001-60	-3.53
M65	N160	10-T5-NS001-60	-2.00
M65	N164A	10-T5-NS002-60	-2.00
M65	N168A	10-T3-NS004-60	-2.65
M65	N168A	10-T3-NS003-60	-4.12
M66	N148	10-T3-NS002-60	-1.41
M66	N148	10-T3-NS001-60	-3.53
M66	N161	10-T5-NS001-60	-2.00
M66	N161	10-T5-NS002-60	-2.00
M66	N169A	10-T3-NS004-60	-2.65
M66	N169A	10-T3-NS003-60	-4.12

Longitudinal Struts

M74	N171	Other	-2.35
M74	N163B	Other	-2.35
M75	N164B	Other	-2.35
M76	N162	Other	-2.35



PROJECT: Propane Dehydrogenation Project
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Calculation No.: PDH-00-ST-CAL-0017
By: Coltin Walsh

Contact No.: A6KV
Date: 30-Jul-2018
Check Level: 2B
Chk'd: JR

BLDG NO	TRAY TAG	TRAY LOAD - kg/m (EXCL. SNOW AND WIND LOAD)	Tray Load [kN / m]	EL for Trib L 6m [kN]	EL for Trib L 4m [kN]
10-BU-78003	10-T3-NS001-60	27	0.26	1.59	1.06
	10-T3-NS002-60	44	0.43	2.59	1.73
	10-T3-NS003-60	26	0.26	1.53	1.02
	10-T3-NS004-60	62	0.61	3.65	2.43
	10-T5-NS001-60	34	0.33	2.00	1.33
	10-T5-NS002-60	34	0.33	2.00	1.33
	Other	60	0.59	3.53	2.35
	Total	287			

Member	Node	Tray Tag	EL [kN]
M79	N173	10-T3-NS003-60	-1.53
M79	N178	Other	-3.53
M79	N183	Other	-3.53
M79	N188	Other	-3.53
M80	N174	10-T3-NS003-60	-1.53
M80	N179	10-T3-NS003-60	-1.53
M80	N179	10-T3-NS001-60	-1.59
M80	N179	10-T5-NS001-60	-2.00
M80	N184	10-T3-NS004-60	-3.65
M80	N184	10-T3-NS002-60	-2.59
M80	N189	Other	-3.53
M80	N189	10-T5-NS002-60	-2.00
M81	N175	10-T3-NS003-60	-1.53
M81	N175	Other	-3.53
M81	N180	10-T3-NS003-60	-1.53
M81	N180	10-T3-NS001-60	-1.59
M81	N180	10-T5-NS001-60	-2.00
M81	N185	10-T3-NS004-60	-3.65
M81	N185	10-T3-NS002-60	-2.59
M81	N185	10-T5-NS002-60	-2.00
M81	N189	Other	-3.53
M82	N176	10-T3-NS003-60	-1.53
M82	N181	10-T3-NS003-60	-1.53
M82	N181	10-T3-NS001-60	-1.59
M82	N181	10-T5-NS001-60	-2.00
M82	N186	10-T3-NS004-60	-3.65
M82	N186	10-T3-NS002-60	-2.59
M82	N186	10-T5-NS002-60	-2.00
M82	N191	Other	-3.53
M83	N182	10-T3-NS003-60	-1.53
M83	N182	10-T3-NS001-60	-1.59
M83	N182	10-T5-NS001-60	-2.00
M83	N187	10-T3-NS004-60	-3.65
M83	N187	10-T3-NS002-60	-2.59
M83	N187	10-T5-NS002-60	-2.00



PROJECT: Propane Dehydrogenation Project
Location: Strathcona County, Alberta
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By: Coltin Walsh

Contact No.: A6KV
Date: 30-Jul-2018
Check Level: 2B
Chk'd: JR

Longitudinal Struts

M86	N191A	Other	-2.35
M86	N192	Other	-2.35

BLDG NO	TRAY TAG	TRAY LOAD - kg/m (EXCL. SNOW AND WIND LOAD)	Tray Load [kN / m]	EL for Trib L 6m [kN]	EL for Trib L 4m [kN]
10-BU-78004	10-T2-PF001-90	34	0.33	2.00	1.33
	10-T2-PF002-90	40	0.39	2.35	1.57
	10-T3-PF001-60	56	0.55	3.30	2.20
	10-T3-PF002-60	51	0.50	3.00	2.00
	10-T3-PF003-60	60	0.59	3.53	2.35
	10-T3-PF004-60	60	0.59	3.53	2.35
	10-T5-PF001-60	28	0.27	1.65	1.10
	10-T5-PF002-60	83	0.81	4.89	3.26
	Other	60	0.59	3.53	2.35
	Total	472			

Member	Node	Tray Tag	EL [kN]
M36	N91	10-T3-PF001-60	-3.30
M36	N91	10-T3-PF002-60	-3.00
M36	N91	10-T5-PF002-60	-4.89
M36	N107	10-T2-PF001-90	-2.00
M36	N107	10-T2-PF002-90	-2.35
M36	N99	10-T3-PF003-60	-3.53
M36	N99	10-T3-PF004-60	-3.53
M36	N99	10-T5-PF001-60	-1.65
M37	N92	10-T3-PF001-60	-3.30
M37	N92	10-T3-PF002-60	-3.00
M37	N108	Other	-1.77
M37	N108	10-T2-PF001-90	-2.00
M37	N108	10-T2-PF002-90	-2.35
M37	N100	10-T3-PF003-60	-3.53
M37	N100	10-T3-PF004-60	-3.53
M37	N100	10-T5-PF001-60	-1.65
M38	N93	10-T3-PF001-60	-3.30
M38	N93	10-T3-PF002-60	-3.00
M38	N111	10-T2-PF001-90	-2.00
M38	N111	10-T2-PF002-90	-2.35
M38	N101	10-T3-PF003-60	-3.53
M38	N101	10-T3-PF004-60	-3.53
M38	N101	10-T5-PF001-60	-1.65
M39	N94	10-T3-PF001-60	-3.30
M39	N94	10-T3-PF002-60	-3.00
M39	N112	10-T2-PF001-90	-2.00
M39	N112	10-T2-PF002-90	-2.35
M39	N102	10-T3-PF003-60	-3.53
M39	N102	10-T5-PF001-60	-1.65
M40	N95	10-T3-PF001-60	-3.30
M40	N113	10-T2-PF001-90	-2.00
M40	N113	10-T2-PF002-90	-2.35

* assumed 3m tributary length only.



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Location: Strathcona County, Alberta
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Calculation No.: PDH-00-ST-CAL-0017
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Date: 30-Jul-2018

Check Level: 2B
Chk'd: JR

M40	N103	10-T3-PF003-60	-3.53
M40	N103	10-T5-PF001-60	-1.65
M41	N96	10-T3-PF001-60	-3.30
M41	N104	10-T3-PF003-60	-3.53
M41	N104	10-T5-PF001-60	-1.65

Longitudinal Struts

M46	N131	10-T5-PF002-60	-3.26
M46	N124A	10-T5-PF002-60	-3.26
M47	N125A	Other	-2.35
M47	N126A	Other	-2.35
M48	N123	Other	-2.35
M48	N122	Other	-2.35
M49	N124	Other	-2.35
M50	N126	Other	-2.35
M50	N127	Other	-2.35



PROJECT: Propane Dehydrogenation Project
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Contact No.: A6KV
Date: 30-Jul-2018
Check Level: 2B
Chk'd: JR

Appendix B
- Force Table -

Inter Pipeline

Alberta PDH Project

Fluor Contract No.: A6KV

PDH-00-ST-CAL-0017_R0

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STRUCTURAL STEEL CONNECTION DESIGN TABLE

EWP No.

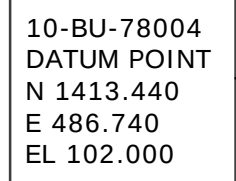
Callout	Sl. No.	Connection Information						Structural Engineer of Record for Connection Design Basis: Fluor							Revision Control	
		Responsibility	Connection		Series	Envelope / Batch	Set	Member Connection Design Force						Notes or Special Instructions	Rev. No.	Status
			Type	Code				Fx (kN)	Fy (kN)	Fz (kN)	Mx (kN-m)	My (kN-m)	Mz (kN-m)			
V1	1	Detailer	Shear	V	01	Envelope	0	25	50	5	-	-	-	Keep 12 clear from edge of pile cap plate or beam flange. Cut beam straight, no cope required.	0	Original
V2	2	Detailer	Shear	V	02	Envelope	0	-	50	5	-	-	-	Axial Slot: 21mm x 70mm long. Keep 35mm clear from edge of pile cap plate or beam flange. Provide PL 10mm washer plate. Cut beam straight, no cope required.	0	Original



PROJECT: Propane Dehydrogenation Project
Location: Strathcona County, Alberta
Title: E&I Building Cable Tray Supports
Calculation No.: PDH-00-ST-CAL-0017
By: Coltin Walsh

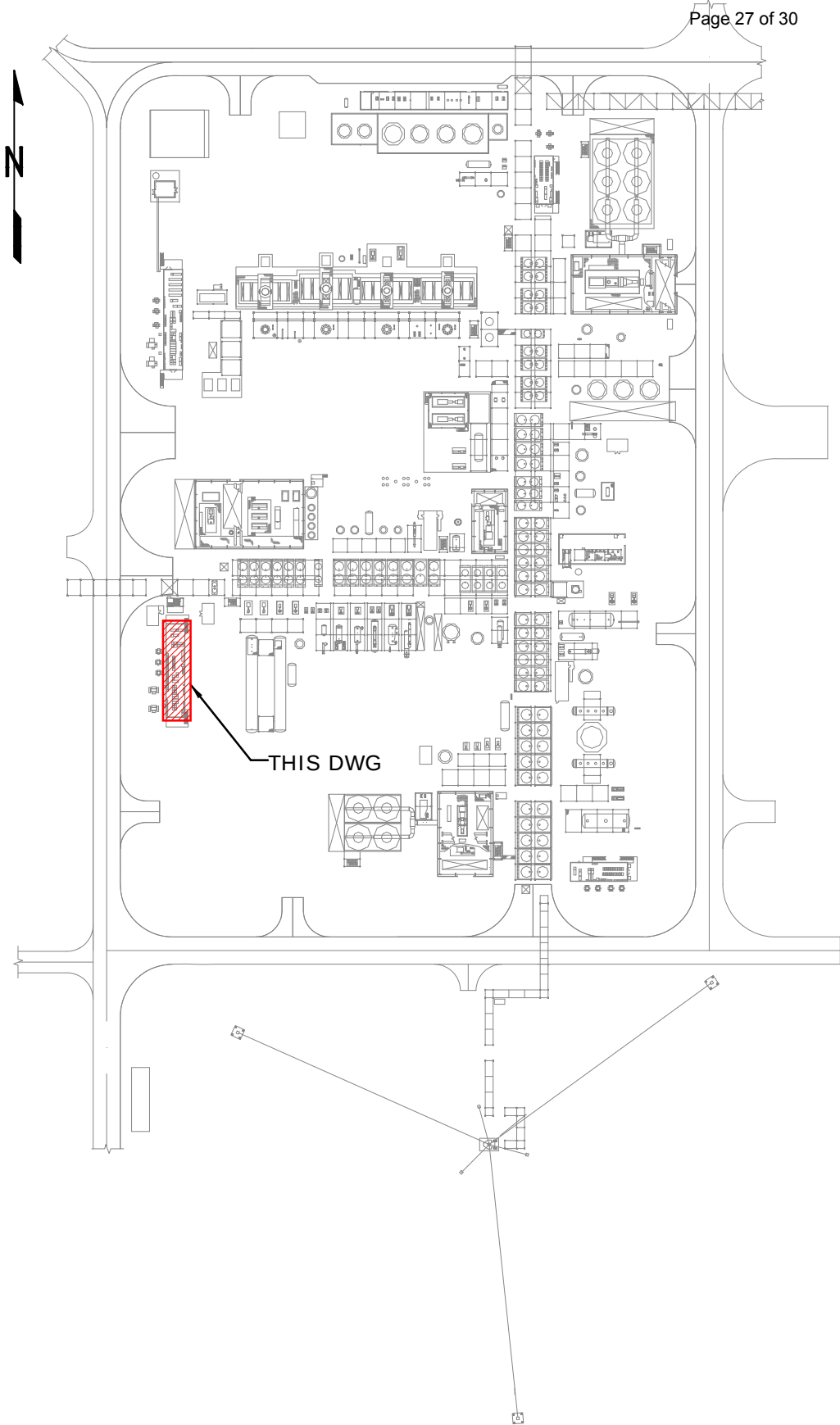
Contact No.: A6KV
Date: 30-Jul-2018
Check Level: 2B
Chk'd: JR

Appendix C - IFC Drawings -



ISOMETRIC VIEW

LOOKING NORTHEAST (DOWN)
SCALE: Fit To Scale



KEY PLAN

NOTES:

1. FOR STRUCTURAL STEEL STANDARDS AND GENERAL NOTES, SEE DRAWING P0H-SD20-00001.
2. CONNECTION CALL-OUTS THAT REFER TO THE STRUCTURAL STEEL STANDARD DRAWINGS USE ABBREVIATED DRAWING NUMBERS.

EXAMPLE: STANDARD CONNECTION DETAIL 5 ON DRAWING P0H-SD20-00014-01 WILL BE SHOWN THUS 14.5
3. REFER TO ASSOCIATED FILES FOR STRUCTURAL MODEL INFORMATION UNLESS NOTED ON DESIGN DRAWINGS.
4. THIS STEEL PACKAGE TO BE DETAILED IN COORDINATION WITH THE PACKAGES LISTED IN THE ADJACENT MODULE INDEX AND ASSOCIATED FIELD ERRECTED INDEX TO ENSURE INTERFACE CONNECTIONS ARE INCORPORATED INTO THE DESIGN.
5. REFER TO VENDOR DRAWINGS FOR INSTALLATION DETAILS.
6. ALL BEAMS CONNECTED TO COLUMN FLANGES SHALL BE DETAILED USING STANDARD CONNECTION DETAIL 9.18 OR 9.19 UNLESS NOTED OTHERWISE.
7. ALL BEAM TO COLUMN WEB CONNECTIONS SHALL USE STANDARD CONNECTION DETAIL 9.14 OR 9.15 AND SHALL BE DESIGNED FOR 50% OF THE MEMBER'S CAPACITY IN COMPRESSION ALONG WITH 25% OF MEMBER'S CAPACITY IN MAJOR AXIS SHEAR TOGETHER WITH A 10 KN OF MINOR AXIS SHEAR APPLIED AT THE TOP FLANGE OF THE SUPPORTED MEMBER UNLESS NOTED OTHERWISE.
8. ALL VERTICAL BRACING END CONNECTIONS THAT ARE LOADED IN COMPRESSION AND ARE NOT PART OF THE LATERAL FORCE RESISTING SYSTEM SHALL BE DESIGNED FOR 50% OF THE MEMBER'S AXIAL CAPACITY IN COMPRESSION.
9. ALL BEAM TO BEAM WEB CONNECTIONS SHALL BE DETAILED USING STANDARD CONNECTION DETAIL 10.4 OR 10.5 AND SHALL BE DESIGNED FOR 50% OF THE MEMBER'S MAJOR AXIS SHEAR CAPACITY TOGETHER WITH A 5 KN MINOR AXIS SHEAR APPLIED AT THE TOP FLANGE OF THE SUPPORTED MEMBER UNLESS NOTED OTHERWISE.
10. ALL PLAN BRACING END CONNECTIONS SHALL BE DESIGNED FOR A 100 KN AXIAL FORCE (TENSION OR COMPRESSION) IN THE BRACE UNLESS NOTED OTHERWISE.
11. MODULE SHALL BE KEPT PROPERLY SUPPORTED EXTERNALLY, UNTIL COMPLETION OF VERTICAL BRACED BAY.
12. CONTRACTOR PERFORMING TRANSPORTATION AND/OR LIFTING OF MODULE SHALL BE RESPONSIBLE FOR DETERMINING THE WEIGHT AND CENTRE OF GRAVITY OF THE MODULE FOR THE PURPOSE OF TRANSPORTATION AND LIFTING.
13. BEAM CONNECTED TO BEAM THROUGH FLANGE SHALL BE SUPPORTED (BOLTED) AT ALL BEAM INTERSECTIONS USING STANDARD CONNECTION DETAIL 10.13 UNLESS NOTED OTHERWISE.
14. THE FORCE TABLE THAT ACCOMPANIES THIS PACKAGE CONTAINS FORCES AT THE TIME OF IFC TO THE FABRICATOR.
FOR ACTUAL AND FINAL FORCES USED IN THE DESIGN OF THE CONNECTIONS, REFER TO STEEL FABRICATOR'S FABRICATION DRAWINGS.



PROPANE DEHYDROGENATION PLANT
STRUCTURAL STEEL ISOMETRIC
STICKBUILT 1002PF103AA
DESIGN AREA 02PF

STRATHCONA COUNTY		ALBERTA	
DRAWN BY	E. CAUTON	SCALE: AS SHOWN	A.F.E.:
CHECKED BY	MZJR	PDH-S29-01203-01 0	
PROJECT ENG.	B.LI		
APPROVED BY			
DATE ISSUED	19-JUL-2018		

NOTES:	REFERENCE DWG.'s			REVISIONS							PERMIT STAMP		ENGINEER STAMP
											PERMIT TO PRACTICE FLUOR CANADA LTD. Signature _____ Date _____ PERMIT NUMBER: P 0778 The Association of Professional Engineers and Geoscientists of Alberta		
	2	PILE AND FOUNDATION LOCATION PLAN-DESIGN AREA 02PF	PDH-CN10-01202										
1	PILE AND FOUNDATION LOCATION PLAN-DESIGN AREA 02PF	PDH-CN10-01201	0	19-JUL-2018	ISSUED FOR CONSTRUCTION - PROJECT 1202	EC	MZ/CW	ES					
NO.	DESCRIPTION	DWG. NO.	NO.	DATE	DESCRIPTION	DRAWN BY	CHECKED BY	PROJECT ENG.	APPROVED BY				

ISOMETRIC KEY

TOP	N/A	-
BTOM	N/A	-
BORTH	N/A	-
BOUTH	N/A	-
BAST	N/A	-
BEST	N/A	-

STICKBUILT EWP	DRAWING NO.
-	-
-	-
-	-
-	-

DRAWING NO.	REV.	TITLE
-	-	-



1. FOR STRUCTURAL STEEL STANDARDS AND GENERAL NOTES, SEE DRAWING PDH-SD20-00001.
2. CONNECTION CALL-OUTS THAT REFER TO THE STRUCTURAL STEEL STANDARD DRAWINGS USE ABBREVIATED DRAWING NUMBERS.

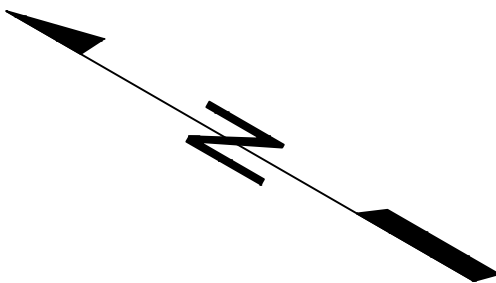
EXAMPLE: STANDARD CONNECTION DETAIL 5 ON DRAWING PDH-SD20-00014-01 WILL BE SHOWN THUS 14.5
3. REFER TO ASSOCIATED FILES FOR STRUCTURAL MODEL INFORMATION UNLESS NOTED ON DESIGN DRAWINGS.
4. THIS STEEL PACKAGE TO BE DETAILED IN COORDINATION WITH THE PACKAGES LISTED IN THE ADJACENT MODULE INDEX AND ASSOCIATED FIELD ERECTED INDEX TO ENSURE INTERFACE CONNECTIONS ARE INCORPORATED INTO THE DESIGN.
5. REFER TO VENDOR DRAWINGS FOR INSTALLATION DETAILS.
6. ALL BEAMS CONNECTED TO COLUMN FLANGES SHALL BE DETAILED USING STANDARD CONNECTION DETAIL 9.18 OR 9.19 UNLESS NOTED OTHERWISE.
7. ALL BEAM TO COLUMN WEB CONNECTIONS SHALL USE STANDARD CONNECTION DETAIL 9.14 OR 9.15 AND SHALL BE DESIGNED FOR 50% OF THE MEMBER'S CAPACITY IN COMPRESSION ALONG WITH 25% OF MEMBER'S CAPACITY IN MAJOR AXIS SHEAR TOGETHER WITH A 10 KN OF MINOR AXIS SHEAR APPLIED AT THE TOP FLANGE OF THE SUPPORTED MEMBER UNLESS NOTED OTHERWISE.
8. ALL VERTICAL BRACING END CONNECTIONS THAT ARE LOADED IN COMPRESSION AND ARE NOT PART OF THE LATERAL FORCE RESISTING SYSTEM SHALL BE DESIGNED FOR 50% OF THE MEMBER'S AXIAL CAPACITY IN COMPRESSION.
9. ALL BEAM TO BEAM WEB CONNECTIONS SHALL BE DETAILED USING STANDARD CONNECTION DETAIL 10.4 OR 10.5 AND SHALL BE DESIGNED FOR 50% OF THE MEMBER'S MAJOR AXIS SHEAR CAPACITY TOGETHER WITH A 5 KN MINOR AXIS SHEAR APPLIED AT THE TOP FLANGE OF THE SUPPORTED MEMBER UNLESS NOTED OTHERWISE.
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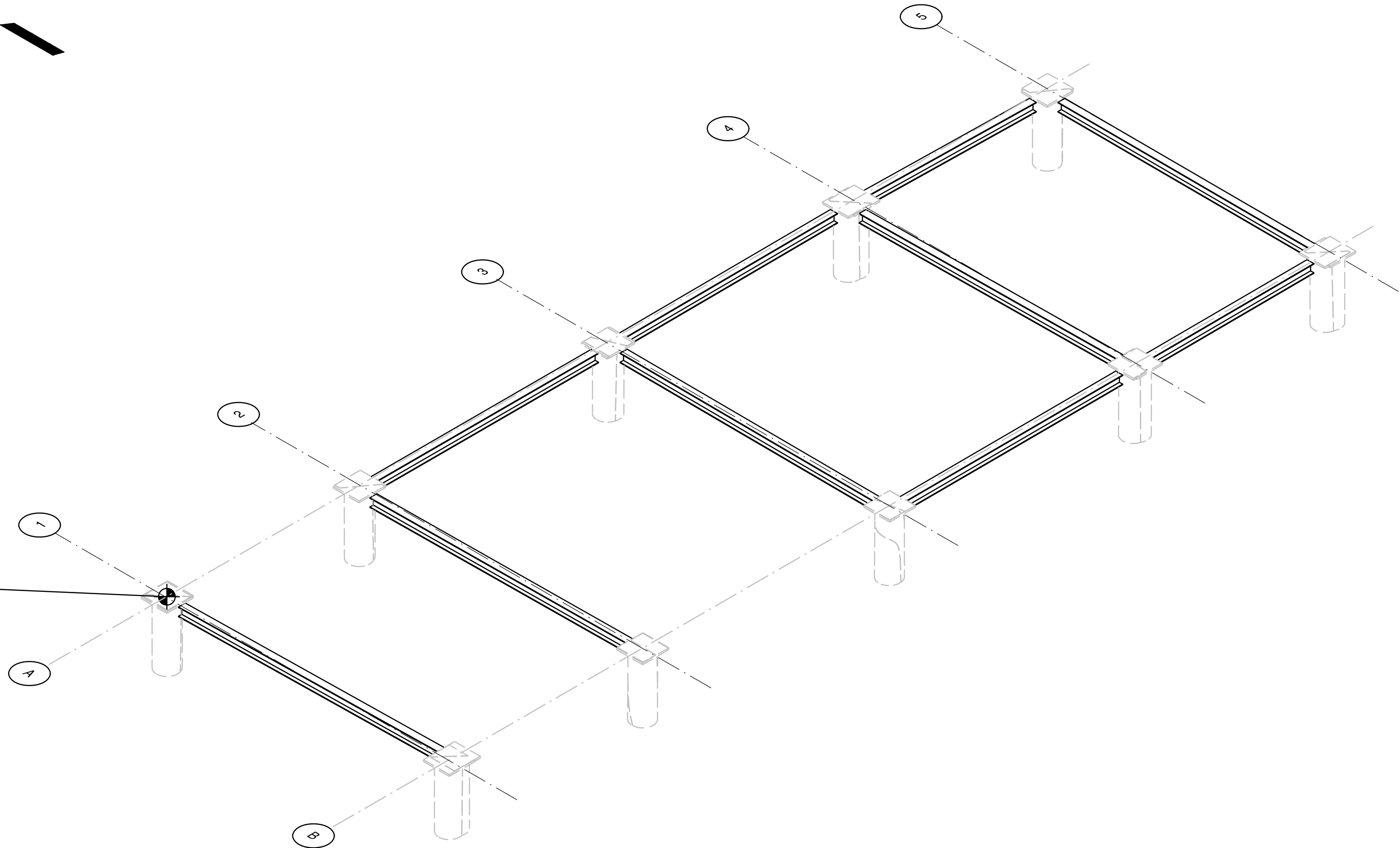
PROPANE DEHYDROGENATION PLANT
STRUCTURAL STEEL ISOMETRIC
STICKBUILT 1001CC106AA
DESIGN AREA 01CC

STRATHCONA COUNTY ALBERTA

DRAWN BY	E. CAUTION	SCALE: AS SHOWN	A.F.E.:
CHECKED BY	RB/JR	DRAWING NO. PDH-S29-06006-01	
PROJECT ENG.	B.LI		
APPROVED BY			
DATE ISSUED	27-JUL-2018		



10-BU-78002
COORD POINT
N 1322.780
E 642.020
EL 102.500



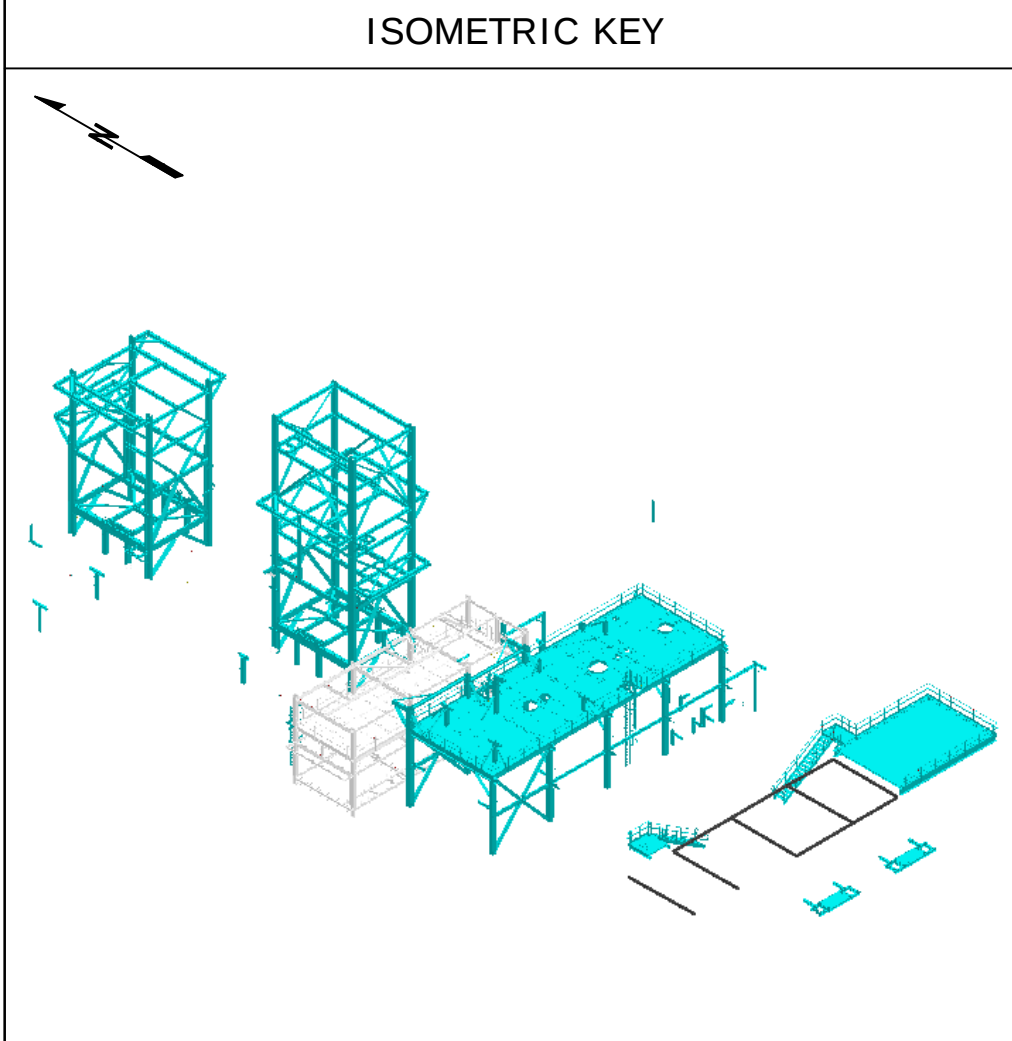
ISOMETRIC VIEW
LOOKING NORTHEAST (DOWN)
SCALE: Fit To Scale

ASSOCIATED FILES	
1002SP107AA_r0.stp	MAIN STRUCTURE CIS/2 FILE
N/A	SHIPPED LOOSE CIS/2 FILE
1002SP107AA_EXPORTSTR_r0.dgn	STRUCTURAL MICROSTATION FILE
1002SP107AA_FORCETABLE_r0.xlsx	CONNECTION FORCES

REFERENCE INFO ONLY	
1002SP107AA_REFERENCE_r0.dgn	REFERENCE ONLY MICROSTATION FILE

LEGEND:

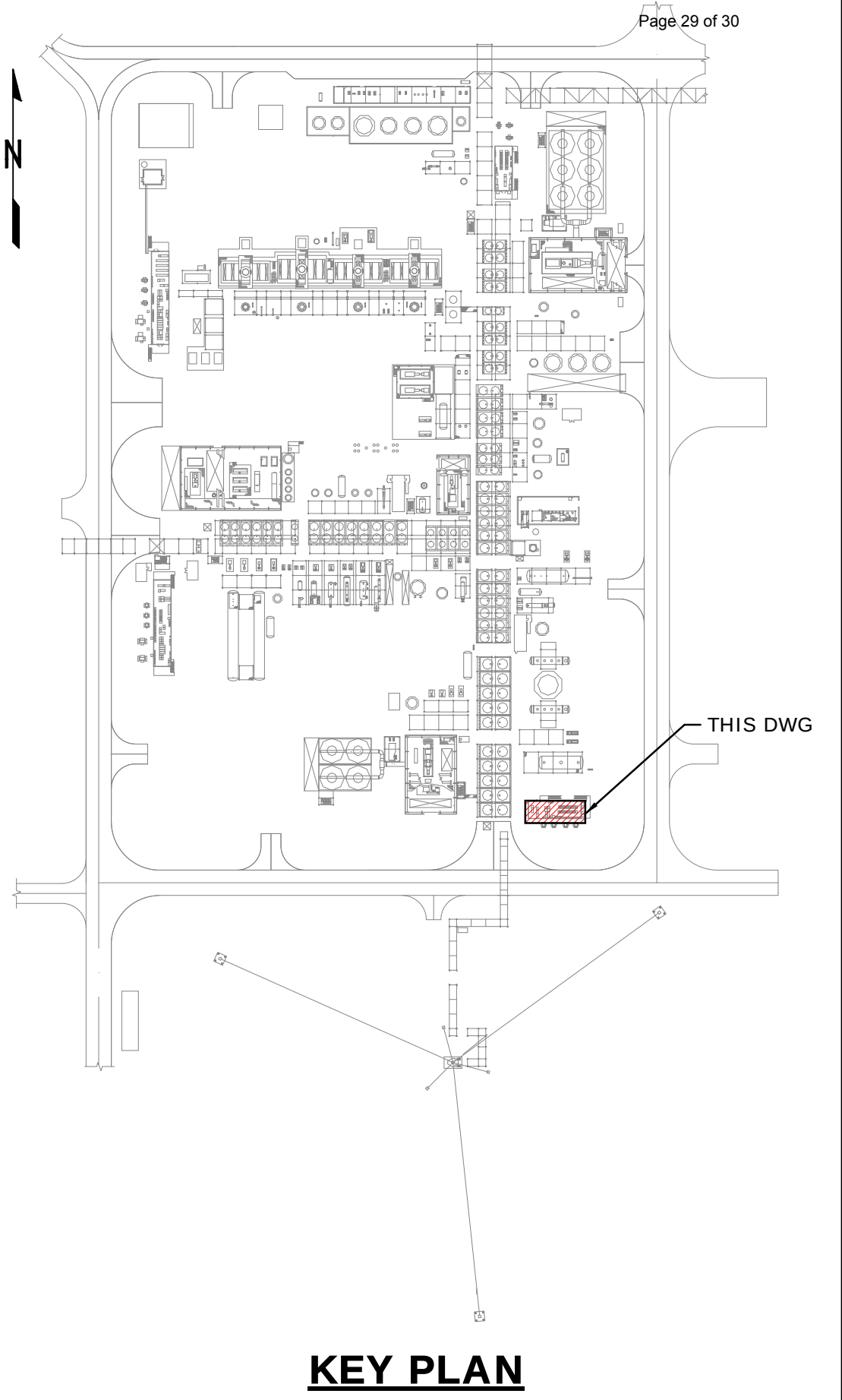
- SHIPPED LOOSE
- STICKBUILT
- MODULAR
- EWP PACKAGE
- CONNECTION CALL-OUT (SEE NOTE 2)



ADJACENT MODULE INDEX		
POSITION	MODULE	DRAWING NO.
TOP	N/A	-
BOTTOM	N/A	-
NORTH	N/A	-
SOUTH	N/A	-
EAST	N/A	-
WEST	N/A	-

ASSOCIATED FIELD ERECTED INDEX	
STICKBUILT EWP	DRAWING NO.
-	-
-	-
-	-
-	-

DRAWING INDEX		
DRAWING NO.	REV.	TITLE
-	-	-



- NOTES:**
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PROPANE DEHYDROGENATION PLANT
STRUCTURAL STEEL ISOMETRIC
STICKBUILT 1002SP107AA
DESIGN AREA 02SP

STRATHCONA COUNTY		ALBERTA	
DRAWN BY	A. M-TAGLE	SCALE: AS SHOWN	A.F.E.:
CHECKED BY	RB/JR	DRAWING NO.	
PROJECT ENG.	B.L.I	PDH-S29-09103-01	
APPROVED BY			
DATE ISSUED	01-AUG-2018	0	

NOTES:									
REFERENCE DWG.'s				REVISIONS					
1	PILE AND FOUNDATION LOCATION PLAN-DESIGN AREA 02SP	PDH-CN10-09102	0	01-AUG-2018	ISSUED FOR CONSTRUCTION - PROJECT 1202	AMT	RB/CW	ES	
NO.	DESCRIPTION	DWG. NO.	NO.	DATE	DESCRIPTION	DRAWN BY	CHECKED BY	PROJECT ENG.	APPROVED BY

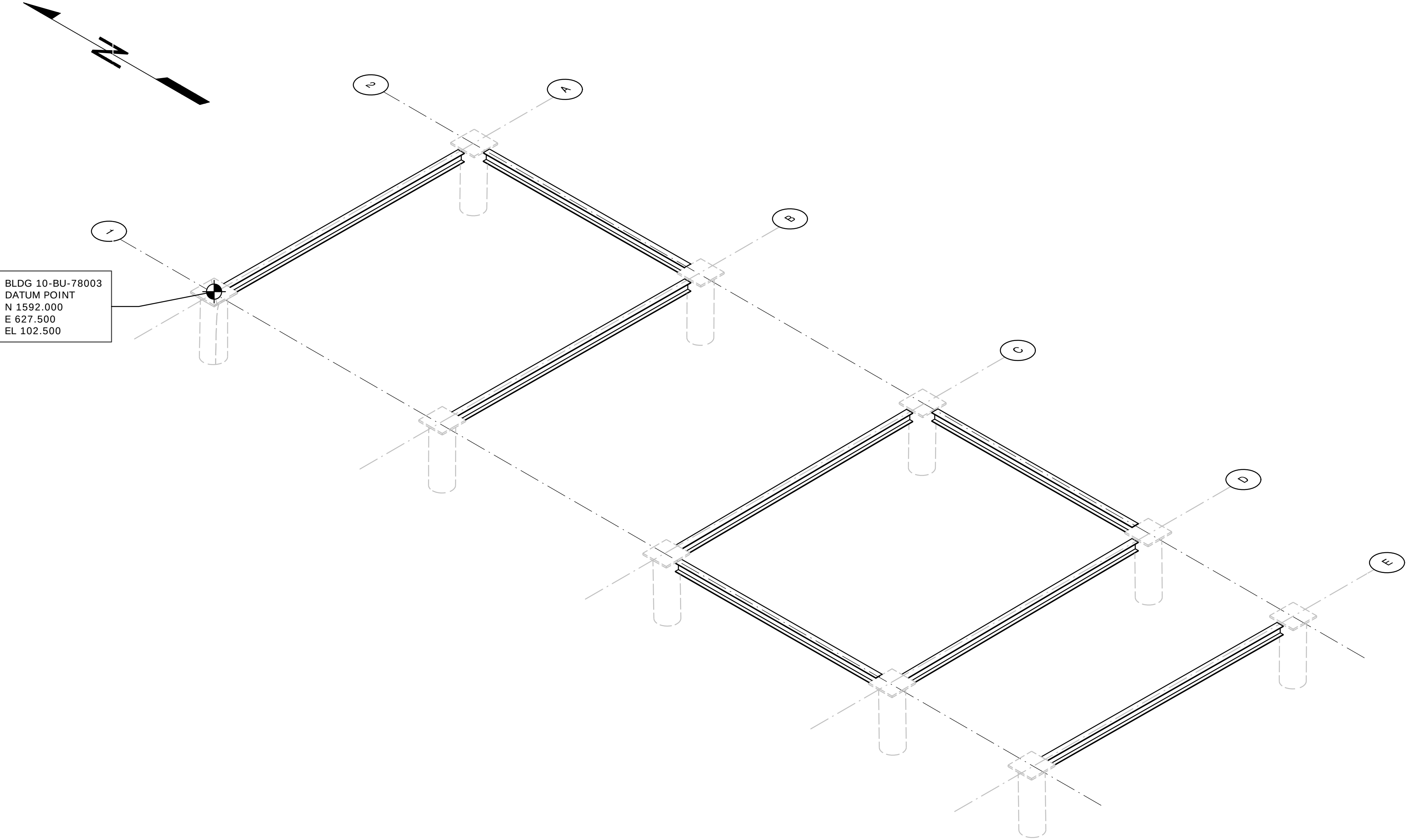
PERMIT STAMP

PERMIT TO PRACTICE
FLUOR CANADA LTD.

Signature _____
Date _____

PERMIT NUMBER: P 0778
The Association of Professional
Engineers and Geoscientists of Alberta

ENGINEER STAMP

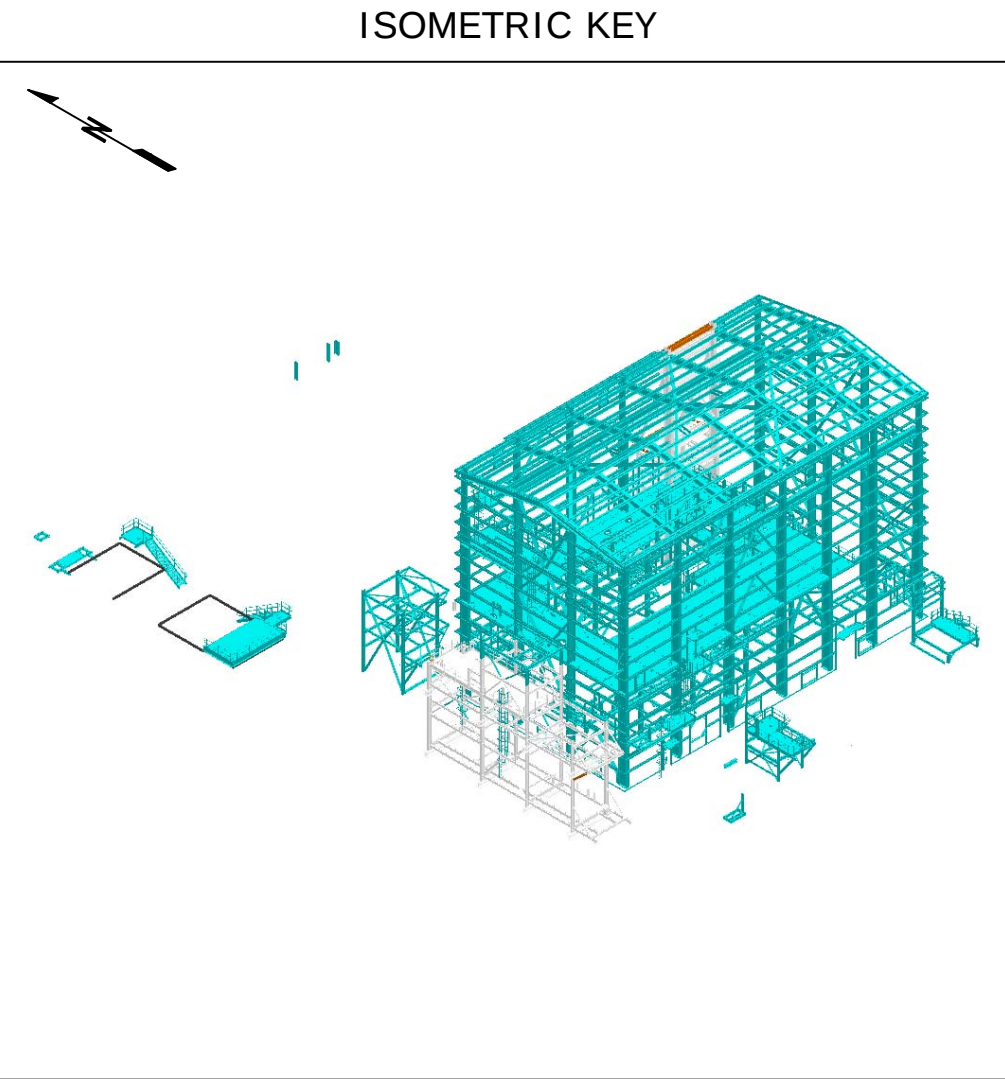


ISOMETRIC VIEW
LOOKING NORTHEAST (DOWN)
SCALE: Fit To Scale

ASSOCIATED FILES	
1001RE109AA_r0.stp	MAIN STRUCTURE CIS/2 FILE
N/A	SHIPPED LOOSE CIS/2 FILE
1001RE109AA_EXPORTSTR_r0.dgn	STRUCTURAL MICROSTATION FILE
1001RE109AA_FORCETABLE_r0.xlsx	CONNECTION FORCES

REFERENCE INFO ONLY	
1001RE109AA_REFERENCE_r0.dgn	REFERENCE ONLY MICROSTATION FILE

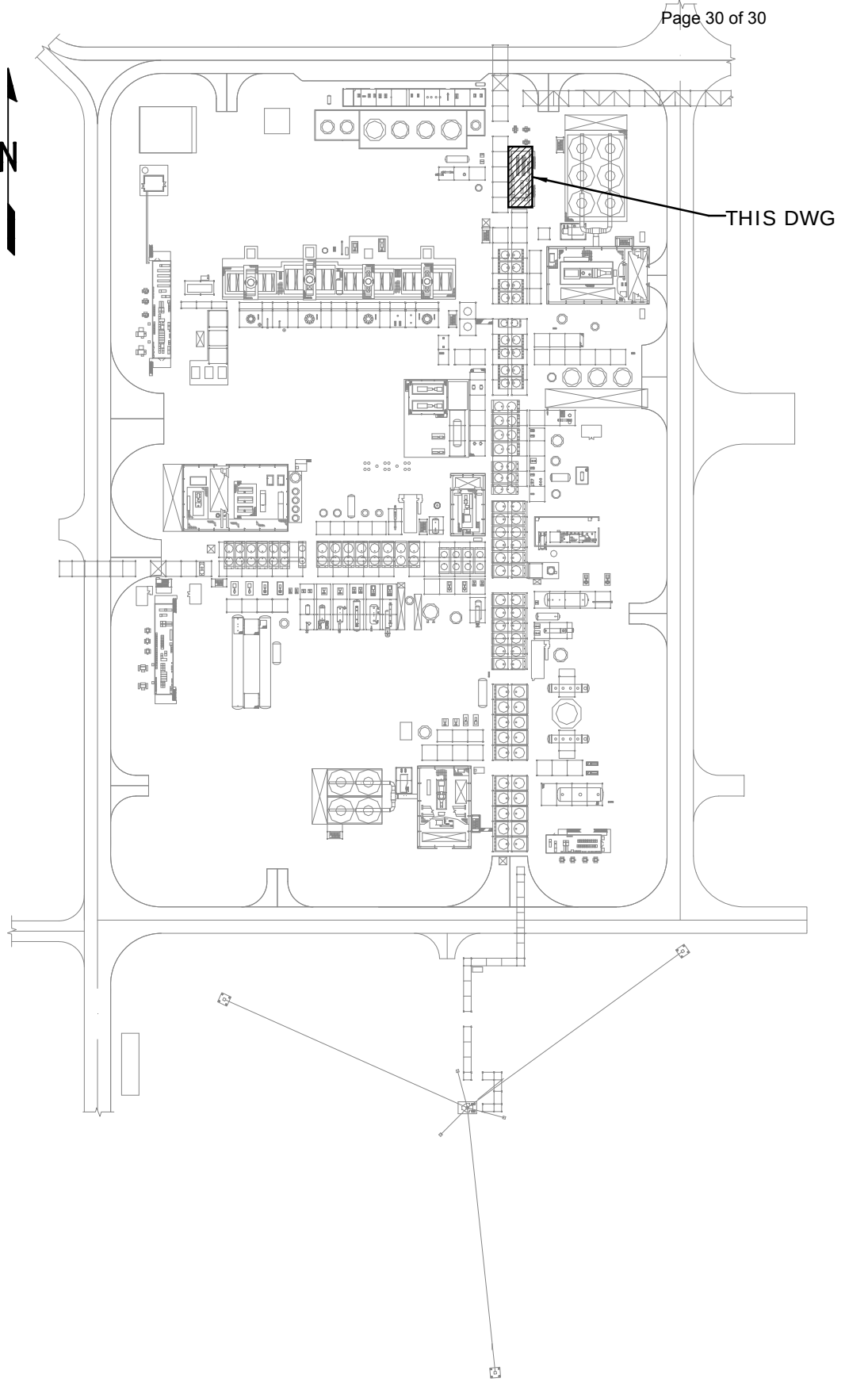
LEGEND:	
	- SHIPPED LOOSE
	- STICKBUILT
	- MODULAR
	- EWP PACKAGE
	- CONNECTION CALL-OUT (SEE NOTE 2)



ADJACENT MODULE INDEX		
POSITION	MODULE	DRAWING NO.
TOP	N/A	-
BOTTOM	N/A	-
NORTH	N/A	-
SOUTH	N/A	-
EAST	N/A	-
WEST	N/A	-

ASSOCIATED FIELD ERECTED INDEX	
STICKBUILT EWP	DRAWING NO.
-	-
-	-
-	-
-	-

DRAWING INDEX		
DRAWING NO.	REV.	TITLE
-	-	-



KEY PLAN

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NOTES:		REFERENCE DWG.'s		REVISIONS						PERMIT STAMP		ENGINEER STAMP	
										PERMIT TO PRACTICE FLUOR CANADA LTD. Signature _____ Date _____ PERMIT NUMBER: P 0778 The Association of Professional Engineers and Geoscientists of Alberta			
3	PILE AND FOUNDATION LOCATION PLAN-DESIGN AREA 03PU	PDH-CN10-12201											
2	PILE AND FOUNDATION LOCATION PLAN-DESIGN AREA 01RE	PDH-CN10-05002											
1	PILE AND FOUNDATION LOCATION PLAN-DESIGN AREA 01RE	PDH-CN10-05001	0	10-AUG-2018	ISSUED FOR CONSTRUCTION - PROJECT 1202	EC	RB/CW	ES					
NO.	DESCRIPTION	DWG. NO.	NO.	DATE	DESCRIPTION	DRAWN BY	CHECKED BY	PROJECT ENG.	APPROVED BY				

PROPANE DEHYDROGENATION PLANT STRUCTURAL STEEL ISOMETRIC STICKBUILT 1001RE109AA DESIGN AREA 01RE			
STRATHCONA COUNTY		ALBERTA	
DRAWN BY	E. CAUTON	SCALE: AS SHOWN	A.F.E.:
CHECKED BY	RB/JR	DRAWING NO.	
PROJECT ENG.	B.L.I	PDH-S29-05008-01	
APPROVED BY			
DATE ISSUED	10-AUG-2018	0	